this case the electric field is not a binding but a disruptive factor. Here, too, there are particles emitted by the nucleus as a result of changes in state, and in this case, we must distinguish between different kinds of particles. First, there are the gamma rays or photons. Analogously to the photons originating in the extranuclear atomic structure, these photons are linked to the electric field in the nucleus. In addition to these, there are the electrons and positrons, emitted in nuclear transmutations, and the neutrinos which always accompany them. The latter are similar to photons in many respects. The only difference is that neutrinos have a spin or angular momentum of $\hbar/2$, whereas that of a photon is either 0 or h.

It seems logical here to assume that the emission of these particles is linked up with the nuclear force field, somewhat in the same way as is the emission of photons with the electric field in the extranuclear atomic structure. But such an analogy would mean that the force between neutron and proton is transmitted because of the electrons, positrons and neutrinos. A similar linking of certain particles to a field must not be misinterpreted as meaning that the field is composed of such particles. The expression ‘composed of’ always suggests that the field might be conceived, to be, as it were, replaced by such particles. Actually, however, field and particles are, so to speak, merely different aspects of the very same concept, as was discussed earlier, in connection with the extranuclear structure of the atom.

The most correct way of expressing the situation is: There is a nuclear field, and in stationary states this nuclear field takes on the aspect of a short-range field, continually diminishing in intensity away from its centre, while in non-stationary processes it takes on the aspect of a wave radiation. The latter can be observed either as a wave radiation or as particles, according to the method of observation employed. We shall attempt to explain this by comparison with the more familiar electric field, by describing the force exerted by one electron upon another in two languages—first, in the language of waves, and then in the language of particles.

We can say, first, that an electron produces an electric field around itself, and that this electric field spreads in conformity with Maxwell’s equations. It may act on another electron and create a force on the latter. The corresponding description in terms of the other aspect is: One electron produces a particle, a photon, and this photon is subsequently absorbed by another electron. Thus in the first phrasing we speak of a ‘production of a field,’ and in the other of a ‘production of a particle’; in the first statement, we refer to an ‘ action of a field’, in the second one to an ‘absorption of a photon by a particle’. This state of affairs can be expressed schematically as follows: